Inventory-based biomass assessment of newly introduced and existing mangrove vegetation for Kappaladi Kite Lagoon restoration project in collaboration with Dilmah Conservation

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Executive Summary

This report presents a comprehensive overview of a multi-phase biodiversity restoration and carbon sequestration initiative led by Dilmah Conservation, combining ecological science, community engagement, and climate action. The project spans terrestrial and coastal ecosystems, contributing directly to Dilmah's 2030 goal of carbon neutrality by offsetting the 75,022 metric tons of CO_2 equivalent (CO_2e) emitted annually by its operations through nature-based solutions.

Fieldwork began in January 2025 with ecological restoration in the Endana Biodiversity Corridor, a fragmented area adjacent to the Sinharaja Rainforest. Activities included species identification, plant height and root collar diameter measurement, herbarium specimen collection, and the development of long-term monitoring systems. Community-based livelihood strategies, such as beekeeping, were incorporated to promote sustainability.

In Kalpitiya, mangrove quadrat sampling was conducted across 10 plots to evaluate species composition, tree health, and biomass. Komiyama's allometric equations were applied to estimate above-ground biomass, with Rhizophora species sequestering up to 10.65 tonnes CO_2/ha/year. Despite an 85% survival rate for planted propagules and strong local participation, the project encountered data limitations, such as the absence of comprehensive environmental datasets and baseline mapping for the entire wetland. These factors, along with high salinity fluctuations and microclimatic variability, introduced uncertainty in biomass estimations and emphasized the need for extended monitoring infrastructure.

Integrated within this initiative is the Kappaladi Kite Lagoon Mangrove Restoration Project, a targeted effort to rehabilitate a degraded mangrove ecosystem in northwestern Sri Lanka. Conducted from February 27–29, 2025, this baseline survey aimed to tag mangrove plants, measure growth characteristics, catalog species, and estimate biomass and carbon sequestration potential. Ten 10m x 10m plots were established across newly planted and existing areas, revealing five mangrove species: *Lumnitzera racemosa*, *Avicennia marina*, *Rhizophora* spp., *Excoecaria agallocha*, and *Aegiceras corniculatum*. The total biomass recorded was 359.26 kg, with plot-level annual CO_2 absorption estimates ranging from 8.96 to 283.47 kg CO_2/year. The average growth condition of newly planted vegetation was recorded at 0.885, while existing mangrove vegetation showed a slightly higher average of 0.894, reflecting promising establishment and health.

To ensure adaptive management and long-term project success, recommendations include periodic remeasurement of tagged plants, continuous biodiversity assessments, and long-term tracking of biomass and carbon storage. The integration of satellite imagery, weather monitoring, and permanent sampling stations is also advised. This initiative represents a scalable model for ecosystem restoration and corporate climate leadership through scientifically grounded, community-inclusive approaches.

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1. Introduction

1. Dilmah Conservation and Its Biodiversity Efforts

but also about long-term **climate resilience and carbon sequestration**.

Collaborations with government agencies like the **Marine Environment Protection Authority (MEPA)** and **National Aquaculture Development Authority (NAQDA)**, as well as with universities and civil society organizations such as *Pearl Protectors*, further enhance the credibility and scalability of this initiative. Plans are also underway to establish a dedicated **research station**, turning the site into a hub for environmental monitoring, education, and replicable restoration methodologies. Dilmah is one of Sri Lanka's most recognized tea brands, built on the founding philosophy of its creator Merrill J. Fernando: *"Business is a matter of human service."* This principle not only guides the company's social mission but also its strong environmental ethic. In 2007, this vision gave rise to **Dilmah Conservation**, the environmental arm of the Dilmah Ceylon Tea Company and the MJF Group. Funded by a portion of global Dilmah tea sales, the organization has grown into a vital force for biodiversity conservation and sustainable development in Sri Lanka.

Dilmah Conservation operates with a mission to integrate **economic sustainability, environmental protection, and community development**. The organization emphasizes that businesses, especially those dependent on natural resources like tea, have a responsibility to safeguard the ecosystems that support them. In this context, protecting biodiversity and sustainably managing natural habitats such as forests, wetlands, and coastal ecosystems are not only ethical imperatives but also long-term investments in resilience.

Through initiatives under the themes of **Sustainability, Biodiversity, Heritage**, and **Education & Awareness**, Dilmah Conservation has launched major conservation projects across Sri Lanka. These include the *Endana Biodiversity Corridor* (EBC), *Greening Batticaloa*, and *DCSARC* (Dilmah Conservation Sustainable Agriculture Research Centre). These projects aim to restore degraded ecosystems, preserve native species, and empower local communities with environmental knowledge. Of particular relevance to this report is the **Kappaladi Kite Lagoon Mangrove Restoration Project**, a long-term effort to restore mangrove ecosystems in the Kalpitiya region.

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2. Introduction to the Kappaladi Kite Lagoon Mangrove Restoration Project

The **Kappaladi Kite Lagoon**, located in the Kalpitiya Peninsula, represents a critically threatened coastal wetland ecosystem. Mangrove degradation in this area has been driven by both natural factors—such as fluctuating salinity, seasonal droughts, and flooding—and human activities like improper waste discharge, agriculture runoff, and encroachment. In response, **Dilmah Conservation initiated restoration efforts in this 25-acre site**, launching replanting activities in 2022 following a baseline survey.

The project emphasizes **community involvement and scientific restoration techniques**. A mangrove nursery has been established with over 800 saplings representing seven native mangrove species. These are nurtured to a suitable growth stage before being transplanted into the degraded zones. Involving local residents in nursery management provides employment while strengthening grassroots engagement. In addition, Dilmah Conservation conducts workshops for schools and environmental organizations to raise awareness about mangrove ecosystems and promote stewardship.

This project also represents a strategic investment in *blue carbon ecosystems*—habitats like mangroves, seagrasses, and salt marshes that are capable of storing large amounts of atmospheric carbon dioxide. Scientific evidence shows that **mangroves can absorb three to five times more CO_2 than terrestrial forests**, making them an essential component of climate change mitigation. As such, the Kappaladi project is not only about habitat recovery

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3. The Importance of Mangrove Ecosystems and the Need for Restoration

Mangroves are unique salt-tolerant trees and shrubs found in tropical and subtropical intertidal zones. They have evolved a range of adaptations to survive in harsh conditions, such as **pneumatophores** (aerial roots), **vivipary** (live seed germination), and **salt-excreting leaves**. These adaptations allow them to thrive where few other plant species can.

The ecological functions of mangroves are critical and multifaceted. Their dense root systems act as **natural barriers**, reducing coastal erosion and buffering communities from storm surges, cyclones, and tsunamis. Mangroves also serve as **nursery grounds** for a wide variety of marine organisms, including fish, prawns, and crabs, many of which support local fisheries. In addition, they offer **nesting habitats for migratory birds** and help maintain ecological balance through nutrient cycling and water purification.

From a climate perspective, mangroves are vital **carbon sinks**, capable of storing carbon for centuries in both their biomass and waterlogged soils. Unlike terrestrial forests, they are less prone to fire and decomposition, making them more effective in long-term carbon storage. Their degradation, however, leads to the rapid release of stored carbon into the atmosphere, exacerbating global warming.

Despite their importance, mangroves have often been misunderstood or undervalued. Many people regard them as swampy wastelands, leading to their destruction for shrimp farming, urban expansion, and agriculture. However, **Sri Lanka has taken significant steps to reverse this trend**, becoming the **first country in the world to legally protect all its mangrove forests in 2015**, as recognized by the United Nations. Through the creation of the *National Expert Committee on Mangrove Conservation and Sustainable Use*, the country has implemented policy frameworks, restoration projects, and community-based initiatives to safeguard these ecosystems.

Figure 1.1: A view of Endana Biodiversity Corrido

Figure 1.2: A view of Kappaladi Kite Lagoon

Objectives

The primary objectives of this study were to identify and tag individual mangrove plants within selected plots located in both newly planted restoration areas and existing natural mangrove stands. Quantitative data on key growth characteristics of the tagged mangrove plants were collected to support further analysis. An initial inventory of the mangrove species present at the restoration site was also created to document biodiversity and species distribution. In addition, data were gathered to enable the subsequent estimation of mangrove biomass and the potential absorption of carbon dioxide, supporting the evaluation of the site's contribution to carbon sequestration.

2. Methodology

The first phase of the biodiversity restoration project was carried out at the Endana Biodiversity Corridor (EBC), focusing on terrestrial forest restoration. Field activities were conducted under the supervision of experts from Dilmah Conservation and the Rajarata University field station, aiming to strengthen practical knowledge in ecological monitoring, species documentation, and community engagement.

2.1 Phase 1: Endana

2.1.1 Plant Measurements

Fieldwork began with the systematic measurement of 120 seedlings planted as part of the corridor restoration. The selected plant species included **Albizia**, **Shorea spp.**, and **Loxococcus rupicola**, all of which are native to the Sinharaja region. For each seedling, we recorded three key growth indicators: **plant height**, **root collar diameter**, and **the number of leaves and branches**. Instruments such as **calipers**, **measuring tapes**, and **manual counters** were used to ensure precision and consistency in measurements. These data serve as a baseline for long-term growth monitoring and help assess plant adaptation and survival in the restored corridor.

Figure 2.1 :measuring tree hight

Figure 2.2: measuring root collar diameter

2.1.2 Herbarium Preparation

A variety of plant samples collected during our hike to the Walankanda Forest Reserve were used to create herbarium specimens. The collected materials were first **pressed using botanical plant presses** and then **dehydrated for 72 hours** using **silica gel** to preserve their structure and prevent decay. Each specimen was then **carefully mounted, labeled with scientific and collection details, and archived** following standard herbarium protocols. These specimens will aid in future species identification and serve as reference materials for ecological studies at the Endana research station.

Figure 2.3 :preparing bundles for dehydrating

Figure 2.4 : dehydrating the processed bundles

2.1.3 Plant tagging

A practical session and lecture on plant tagging methods relevant to restoration projects was conducted. This included instruction on the preparation of durable aluminium tags using numbered and lettered blocks and a hammer. Tags that are needed for mangrove restoration project were prepared by us.

Figure 2.5 :final a tag

Figure 2.6 :making tags

2.1.4 Community Integration

The Endana biodiversity restoration initiative actively involved the local community in activities such as beekeeping, nursery management, and seed collection. These efforts promoted community ownership and provided economic benefits. Environmental education sessions were also conducted to raise awareness. The project successfully combined conservation with local livelihood support, ensuring long-term sustainability.

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Figure 2.7 :subcultures of the common people

Figure 2.8 :on observation tour of the area

2.1.5 Experience Plant nursery & manage project finance

During the visit to the plant nursery, we engaged in an in-depth discussion on the selection of suitable native plant species for ecological restoration efforts. This was followed by hands-on experience in preparing potting mixtures and sowing seeds, providing practical insights into nursery-level propagation techniques.

A valuable discussion was also held with Mr. Amila Perera, focusing on the financial management aspects of conservation projects and the application of statistical methods in analyzing ecological data.

The day concluded with a commemorative tree planting event at the Endana Research Station. Native species such as *Loxococcus rupicola*, *Shorea oblongifolia*, *Dipterocarpus* sp., and *Mangifera zeylanica* were planted as part of this symbolic and ecologically significant activity.

Figure 2.9 :study plant nursery

Figure 2.10 :filling the soil mixture for planting packets

2.2 Phase 2: Kalpitiya – Mangrove Restoration and Carbon Assessment

Figure 2.11 :A picture of a mangrove

The second phase of the biodiversity restoration project was carried out in **Kalpitiya**, a coastal region characterized by degraded mangrove habitats, fluctuating salinity levels, and tidal exposure. The objective was to assess the carbon sequestration potential of restored mangrove plots and to apply rigorous ecological monitoring to measure biomass accumulation, tree health, and ecosystem recovery. Mangroves are globally recognized for their high carbon storage capacity, especially in tropical coastal zones, making this phase vital for evaluating nature-based climate mitigation solutions.

2.2.1 Quadrat Design and Site Selection

To ensure representative sampling across the mangrove ecosystem, we established **ten 10m × 10m quadrats** in locations that varied in environmental conditions—particularly **salinity gradients and tidal exposure**. Each quadrat was georeferenced using a **handheld GPS unit**, allowing for accurate mapping, future relocation, and landscape-level GIS analysis. The sites were categorized into zones (e.g., Zone A, B, and C), with **Zone A** representing low salinity and **Zone C** representing high salinity and tidal influence. This zoning enabled a comparative study of mangrove performance under different stress conditions.

The quadrat method was selected due to its scientific robustness for vegetation studies. It allowed consistent measurement of tree density, species distribution, and structural characteristics in manageable, replicable plots. Quadrat sampling also supports integration with remote sensing data and facilitates long-term ecological monitoring programs.

Figure 2.12 :setting up stakes to mark boundaries

2.2.2 Field Data Collection

Each individual mangrove tree within the quadrats was tagged and evaluated. Three key biometric parameters were recorded for every tree:

- **Diameter at Breast Height (DBH)** – measured using diameter tape at 1.3 meters above the ground or prop root point.
- **Tree Height** – measured using clinometers or visual range estimation, validated through staff rods.
- Plant Species Identified:

  Across all surveyed plots, five mangrove species were identified. Including,

  1. Lumnitzera racemosa (Beriya)

  2. Avicennia marina (Manda)

  3. Rhizophora sp. (Kadol)

  4. Excoecaria agallocha (Thela Keeriya)

  5. Aegiceras corniculatum (Heen Kadol/ Averi Kadol)

In addition to tree-level data, we also collected **soil salinity readings** using handheld refractometers and **soil depth measurements** using augers and meter rods. Tree condition (healthy, stressed, or damaged) was noted to support ecological health assessments.

This integrated dataset provided the foundation for accurate **biomass calculation**, species-specific performance analysis, and mapping of survival rates under varying environmental stressors.

Figure 2.12 :data collecting for mangrove

2.2.3 Biomass and Carbon Estimation Models

To estimate the **above-ground biomass (AGB)** of mangrove trees, we applied the allometric equation developed by **Komiyama et al. (2005)**, which is widely recognized in tropical forest biomass research and is tailored for mangrove species:

AGB (kg)=0.251×(ρ×D2×H)0.916\text{AGB (kg)} = 0.251 \times (\rho \times D^2 \times H)^{0.916}AGB (kg)=0.251×(ρ×D2×H)0.916

Where:

- DDD = DBH in centimeters
- HHH = tree height in meters
- ρ\rhoρ = wood density in g/cm^3 (species-specific)

This equation accounts for the variation in tree form and species-specific structural properties, providing more accurate biomass estimates than general forest formulas.

For **below-ground biomass (BGB)**, we applied the **IPCC (2006)** standard conversion ratio of **0.24 × AGB**, which reflects typical root biomass proportions for mangrove ecosystems.

Carbon and CO_2 Equivalence

To derive **carbon content**, we followed standard assumptions that **carbon constitutes 50% of dry biomass**. This is consistent with global guidelines for forest carbon accounting.

For **CO_2 sequestration estimates**, we used the IPCC multiplier:

CO$_2$ (kg)=Total Biomass (kg)×1.83\text{CO$_2$ (kg)} = \text{Total Biomass (kg)} \times 1.83CO$_2$ (kg)=Total Biomass (kg)×1.83

This factor converts elemental carbon mass into its CO_2 equivalent based on molecular weight ratios. This calculation allowed us to estimate the amount of atmospheric carbon dioxide absorbed by each mangrove tree per year.

Oxygen (O_2) Production

In addition to carbon sequestration, mangroves contribute to **oxygen generation** through photosynthesis. Using ecological literature (Nowak et al., 2007), we applied the following average ratio:

O_2 produced (kg/year)≈Biomass×1.2\text{O_2 produced (kg/year)} ≈ \text{Biomass} × 1.2O_2 produced (kg/year)≈Biomass×1.2

While approximate, this metric provides an additional lens through which to assess the **ecosystem service value** of restored mangroves beyond just carbon offsets.

2.2.4 Tree Condition Assessment Methodology

Step 1: Individual Tree Assessment

1. **Visit each tree** within your study area.
2. **Observe and record** tree condition based on visible health, canopy state, structural integrity, and signs of damage or decay.
3. **Assign a condition factor** as follows:
   - **A = 1.0** – Good quality, stable, healthy canopy
   - **B = 0.9** – Minor defects or minor damage
   - **C = 0.75** – Major defects, stem damage, lacks stability
   - **D = 0.5** – Dying, severely defoliated, stem decay

Step 2: Plot-Level Analysis

1. Divide the area into **distinct plots**.
2. For each plot:
   - Calculate the **average condition factor**
   - Record this average for comparisons across plots.

Step 3: Group Classification – Newly Planted vs Existing Trees

1. Categorize trees into:
   - **Newly Planted**: Trees planted during the current restoration project.
   - **Existing Planted**: Trees planted in earlier phases or previous years.
2. For each group:
   - Record individual condition factors.
   - Compute **average condition factor per group**

Step 4: Comparative Analysis

1. Compare:
   - **Average condition between plots** to identify differences in tree health across the landscape.
   - **Average condition of newly planted vs. existing trees** to evaluate the success and health of recent planting efforts.

3. Results

3.1 Lessons Learned from the Endana Program for the Kalpitiya Restoration Project

The most significant outcome of participating in the **Endana restoration program** was that it gave us the **practical knowledge and confidence** to independently carry out our own project in **Kalpitiya**. The experience was hands-on and deeply educational, preparing us in several important ways:

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1. Field Plotting and Data Collection

At Endana, we identified and plotted multiple reforestation sites and received hands-on training in essential plant data collection. We learned how to measure Diameter at Breast Height (DBH) and plant height—skills that proved crucial for our later work in Kalpitiya, where we applied the same methods to collect accurate baseline data.

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2. Tag Making and Tree Identification

We were taught how to make and label tree tags, an important skill for monitoring and long-term ecological studies. We created the actual tags used in the Kalpitiya project, which helped us track species, growth, and plant condition over time.

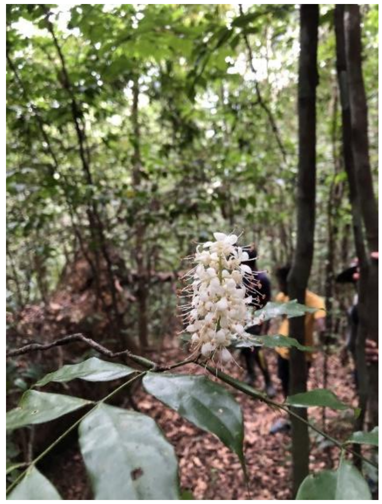

Figure 3.1 :tree identification during the walankanda hike

2. Species Selection Based on Site Conditions

We learned how to select suitable tree species for reforestation by considering factors such as soil type, topography, sunlight exposure, and the presence of existing native vegetation. This knowledge helped ensure ecological compatibility and improved the survival rates of planted species in Kalpitiya.

Figure 3.2 :in a moist ,rocky soil

Figure 3.3 :on a sloping ground

1. **Nursery Work and Seed Collection**

We actively collected seeds from different areas of the forest, ensuring genetic diversity even within the same species to enhance plant resilience. We also helped nurture these seedlings in nurseries until they were ready for field planting.

2. **Independent Application**

All of these skills—data collection, tagging, species selection, and nursery management—prepared us to take full ownership of the Kalpitiya mangrove restoration project. We weren't just assisting; we were leading, empowered by the Endana program with the knowledge, techniques, and confidence to conduct restoration work independently and scientifically.

3. **Herbarium Sheet Preparation**

We learned how to prepare herbarium sheets, which are essential for documenting and preserving plant specimens for future research. By collecting, pressing, drying, and mounting plants, and recording key scientific details such as species name, location, date, and habitat, we deepened our understanding of plant identification and the value of long-term biodiversity documentation.

3.2 Kalpitiya Findings – Mangrove Restoration

1. Establishment of Mangrove Plot

A 10×10 meter mangrove plot was successfully established in Kalpitiya using durable, paint-coated wooden poles to clearly mark the boundaries and provide long-term visibility and protection.


Figure 3.4 :measuring 10m*10m plot


Figure 3.5 :looking for locations covering all zones for plot

3. Tree Tagging and Data Collection

In the next phase, individual mangrove trees within the plot were tagged, and data collection was initiated to support the development of a comprehensive Kalpitiya Mangrove Inventory.


Figure 3.6 :taking measurement


Figure 3.7 :taking measurements of well-grown plant

3. Inventory Management and Digital Records

Collected data—such as species, DBH, and height—were organized into an Excel spreadsheet, creating a digital inventory that serves as a long-term tool for monitoring growth, survival, and environmental impact.

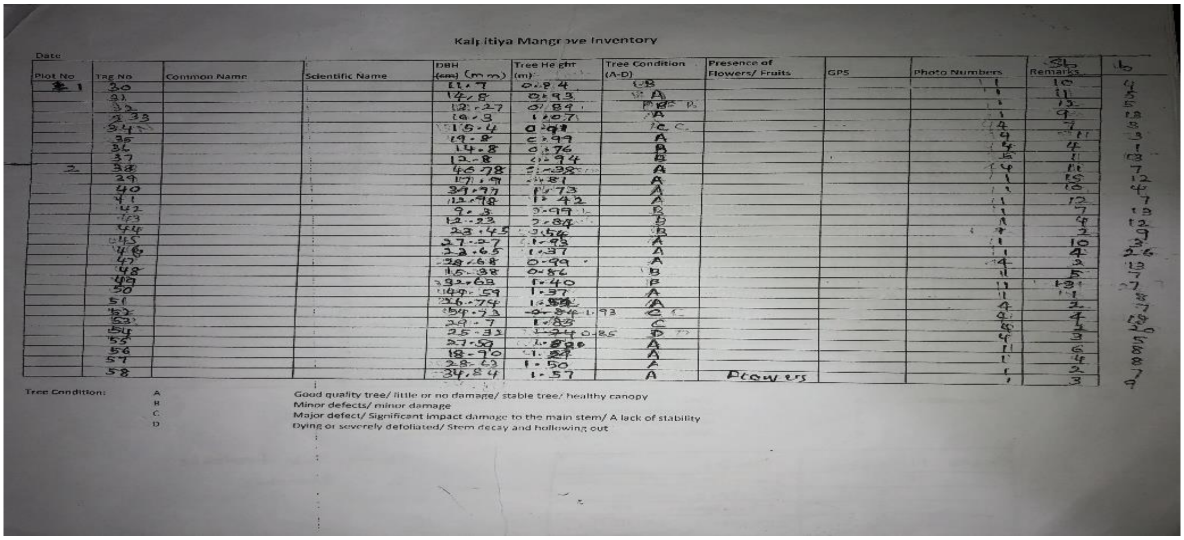

Figure 3.8 :one page of the inventory we created using the data we obtained

Table 3.1:a section of one page of the digital inventory created using inventory

Plot No	Tag No	Common Name	Scientific Name	DBH(mm)	Tree Height(m)	Tree Condition(A-D)
10	311	තෙලකිරිය	*Excoecaria agallocha*	41.16	0.92	A
	312	තෙලකිරිය	*Excoecaria agallocha*	46.39	1.02	B
	313	තෙලකිරිය	*Excoecaria agallocha*	30.88	0.88	B
	314	තෙලකිරිය	*Excoecaria agallocha*	32.88	0.93	B
	315	තෙලකිරිය	*Excoecaria agallocha*	54.31	1.25	B
	316	තෙලකිරිය	*Excoecaria agallocha*	45.91	1.35	B
	317	තෙලකිරිය	*Excoecaria agallocha*	41.41	1.1	A
	318	තෙලකිරිය	*Excoecaria agallocha*	45.48	1.1	A
	319	තෙලකිරිය	*Excoecaria agallocha*	43.31	1.15	A
	320	තෙලකිරිය	*Excoecaria agallocha*	43.16	0.78	B

321	තෙලකීරිය	*Excoecaria agallocha*	26.76	0.73	B
322	තෙලකීරිය	*Excoecaria agallocha*	27.47	1.06	B
323	තෙලකීරිය	*Excoecaria agallocha*	34.5	1.3	A

4. Biomass and Carbon Analysis

Biomass calculations were conducted using standard allometric equations specific to the tree species present in each plot. Based on these biomass values, the estimated amount of CO_2 sequestered by each plot was determined. The average CO_2 sequestration across the 10 plots was found to be 10.65 tonnes of CO_2 per hectare per year. Additionally, the total adjusted above-ground biomass (AGB) was calculated to be 359.26 kilograms per year. These findings demonstrate the effectiveness of the mangrove restoration efforts in enhancing carbon sequestration and contributing to climate change mitigation through increased biomass accumulation.

5. Plot-wise Summary Reports

Using the obtained DBH, Height, and wood density, the AGB value was calculated from mangrove tree to tree, and the adjusted AGB value was obtained based on the obtained tree condition, and the total value was obtained from plot to plot.

Table 3.2 :plot wise summery total adjested AGB

Plot	Total Adjusted AGB (Kg/plot)
1	17.37353951
2	67.38745945
3	12.7776987
4	4.004506622
5	124.9214078
6	14.70627838
7	98.03361567
8	3.95039126
9	9.117384573
10	6.984499435

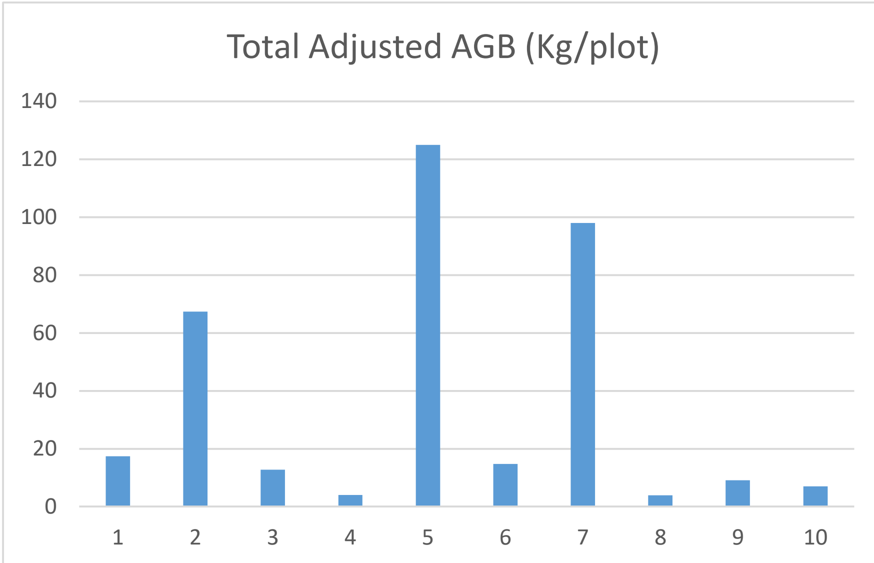

Figure 3.9 :the bar graph of total adjusted AGB in plot 10

6. Area-Based Estimation

Using the total area of each plot, the team scaled up the carbon absorption data. This approach enabled a more accurate extrapolation of the carbon sequestration potential across the entire study area.

Table 3.3 :plot wise CO_2 Sequestered

Plot	CO_2 Sequestered (kg/year)	CO_2 Sequestered (tonnes/ha/year)
1	39.42403585	3.942403585
2	152.915623	15.2915623
3	28.99515389	2.899515389
4	9.087026427	0.908702643
5	283.4716585	28.34716585
6	283.4716585	28.34716585
7	222.4578807	22.24578807
8	8.964227848	0.896422785
9	20.68916907	2.068916907
10	15.84922612	1.584922612

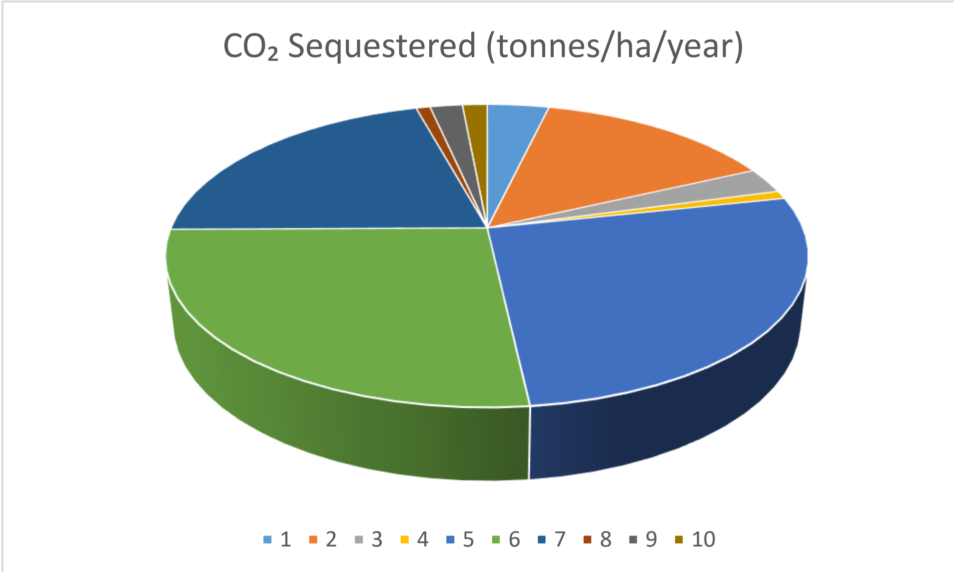

Figure 3.10 :the pie graph of CO_2 sequestered for plots

7. We calculated plot wise tree condition

The average growth condition of newly planted mangrove vegetation was recorded at 0.885, while the existing mangrove vegetation showed a slightly higher average growth condition of 0.894. This indicates that, although the newly planted mangroves are performing well, the established vegetation maintains a marginally better overall growth condition.

Table 3.4 :plot wise average tree condition (yellow color :existing planted ,blue :newly planted)

plot NO	Average Tree Condition value
1	0.933783784
2	0.928571429
3	0.638709677
4	0.891666667
5	0.890350877
6	0.9
7	0.897826087
8	0.939583333
9	0.905128205
10	0.94375

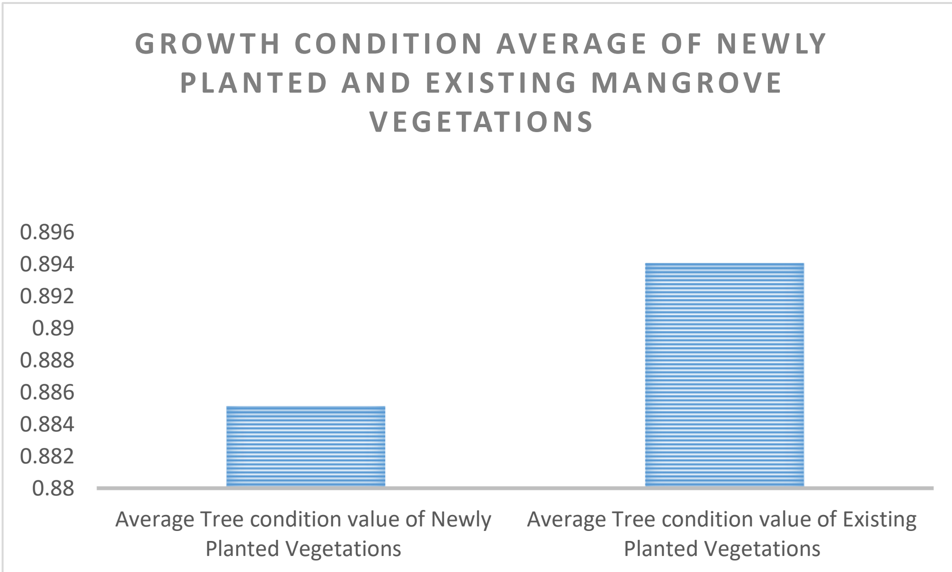

Figure 3.11 :the bar graph average tree condition value of newly planted Vs existing planted

9. Standardize tree condition for every plant (plot wise)

The graph obtained using the standardized value after calculating the average and standard deviation based on the tree condition value obtained from the plants in a plot is shown below.

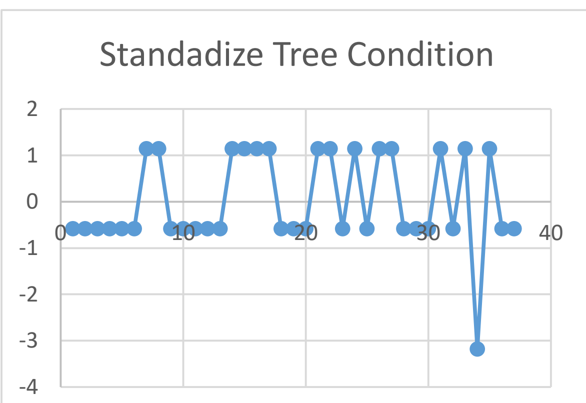

Figure 3.12:standardize tree condition plot 1

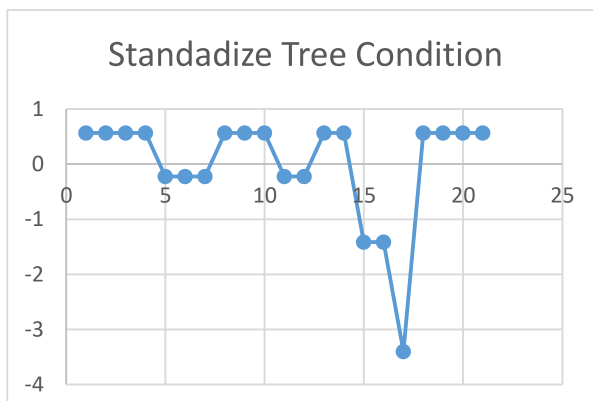

Figure 3.13:standardize tree condition plot 2

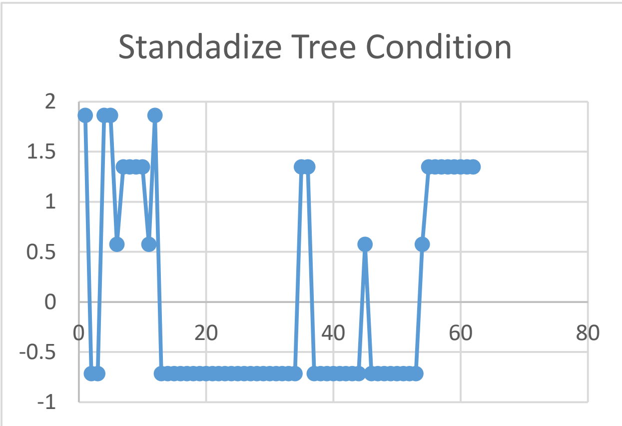

Figure 3.14:standardize tree condition plot 3

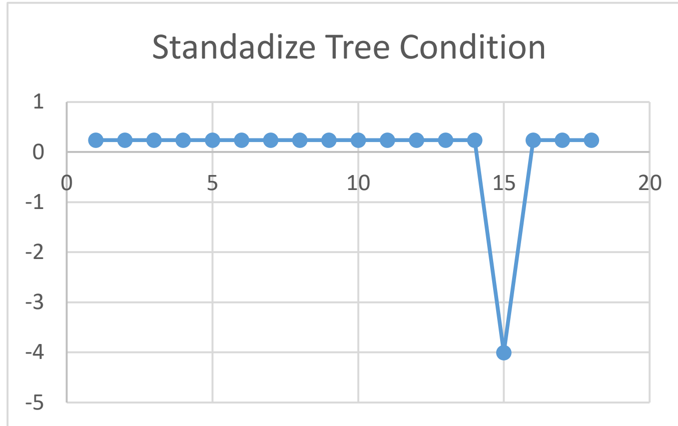

Figure 3.15:standardize tree condition plot 4

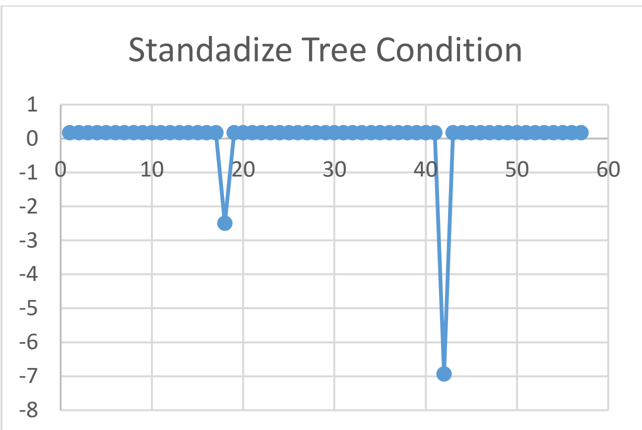

Figure 3.16:standardize tree condition plot 5

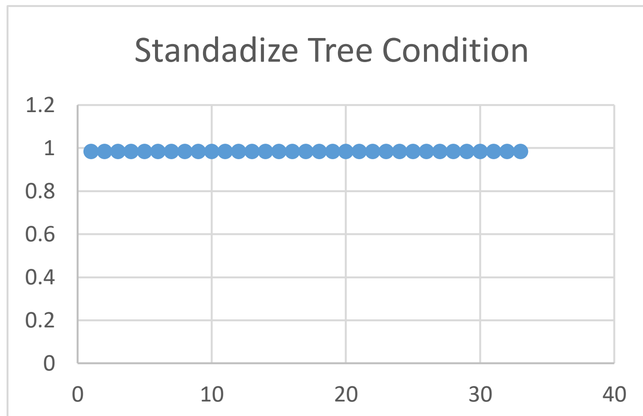

Figure 3.17:standardize tree condition plot 6

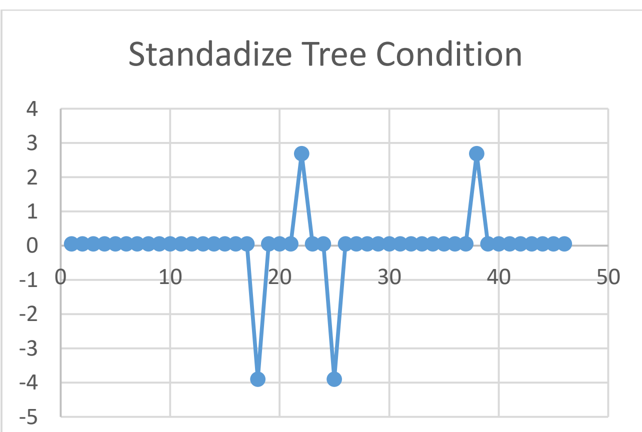

Figure 3.18:standardize tree condition plot 7

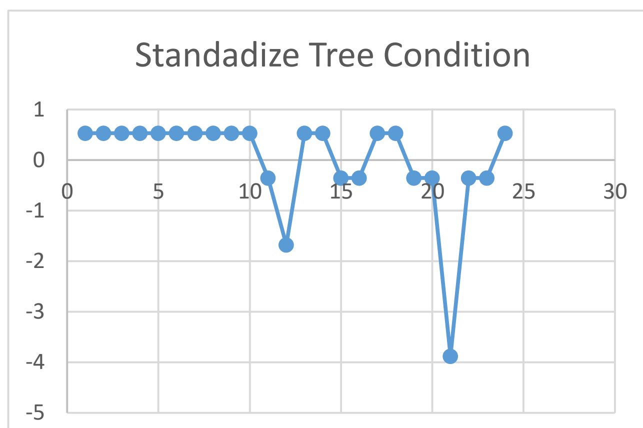

Figure 3.19:standardize tree condition plot 8

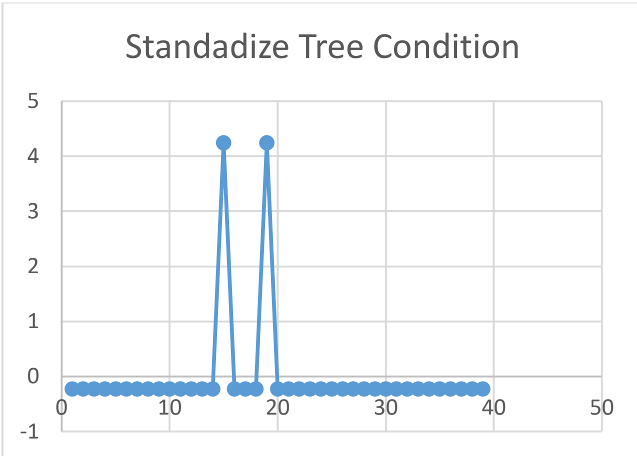

Figure 3.20:standardize tree condition plot 9

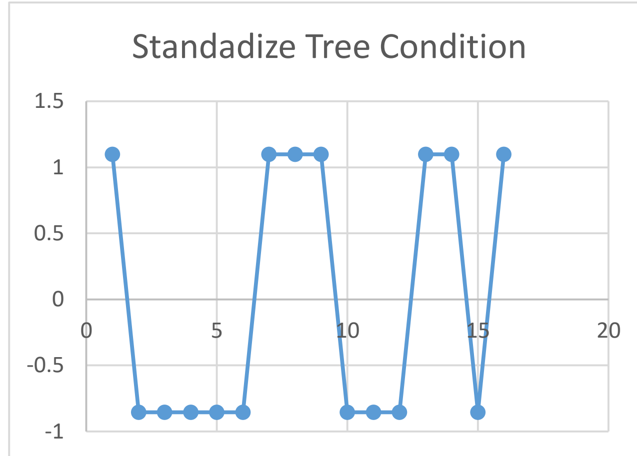

Figure 3.21:standardize tree condition plot 10

Final Analysis Output – Kalpitiya Mangrove Restoration

- **Average CO_2 Sequestration Across 10 Plots:**
  10.65 tonnes of CO_2 per hectare per year

This value is based on detailed **biomass data collection and analysis** conducted within the 10×10 meter plots across the Kalpitiya mangrove restoration site. The calculation reflects the **carbon absorption capacity** of the restored mangrove ecosystem and highlights its significant **climate change mitigation potential**.

This final result can serve as a **baseline for future ecological monitoring**, support **carbon offset initiatives**, and demonstrate the **ecological and environmental value** of mangrove conservation in Kalpitiya.

- **Importance of the Kalpitiya Mangrove Inventory:**
  The **Kalpitiya Mangrove Inventory** created through this project is a **valuable long-term dataset**. As data is collected again after **one year or multiple years**, the inventory can be **updated and expanded** to track changes in:
    - Tree growth and survival
    - Biomass accumulation
    - Carbon sequestration trends
    - Ecological succession

Over time, this updated inventory will become an **essential scientific and conservation tool**, helping to guide **future restoration efforts**, assess the **effectiveness of interventions**, and provide **evidence for sustainable environmental planning**.

4.Discussion

- The Kalpitiya Mangrove Restoration Project represents a successful and practical application of the theoretical and hands-on training received during the Endana Biodiversity Restoration Program. Knowledge and field techniques acquired at Endana empowered students to plan, implement, and scientifically monitor restoration activities in Kalpitiya with a high degree of independence and professionalism. The transition from a guided training environment in Endana to a student-led project in Kalpitiya demonstrates the effectiveness of immersive field-based education in building ecological and scientific competency.
- At Endana, students were introduced to essential forest restoration skills, including site selection, plot marking, and data collection techniques such as measuring tree diameter at breast height (DBH) and plant height. These skills were foundational in enabling the team to accurately assess tree growth and biomass. In Kalpitiya, participants applied these techniques by establishing ten 10x10 meter mangrove plots under various environmental conditions. They installed durable markers, painted plot boundaries, and tagged each tree using methods modeled after Endana. These systematic tagging practices ensured consistent monitoring and data collection, which is critical for long-term ecological assessment.
- Drawing inspiration from the Endana program, students developed the Kalpitiya Mangrove Inventory—a digital database recording species identification, DBH, plant height, and condition scores. Organized using Excel, the inventory supports efficient tracking of growth, mortality, and species distribution over time. This digital tool has long-term value, providing a baseline for continued monitoring and adaptive management. It could also support research on climate resilience, biodiversity offsets, and national conservation strategies.
- One of the project's significant outcomes was the calculation of carbon dioxide (CO_2) sequestration, which averaged 10.65 tonnes per hectare per year. This was determined using species-specific allometric equations introduced during the Endana training. Students applied these methods confidently, calculating aboveground biomass and estimating the ecosystem's contribution to climate change mitigation. These findings highlight the global ecological relevance of localized restoration efforts. Given the role of mangroves as efficient carbon sinks, this CO_2 absorption benchmark can be used to support future carbon offset proposals and attract policy and funding attention to community-based restoration work.
- The nursery training received at Endana also played a crucial role in Kalpitiya's success. Students were taught to collect seeds from a wide genetic pool within each species,

enhancing resilience against environmental stressors like salinity, pests, and storms. This genetically diverse stock was cultivated using techniques such as controlled watering, shading, and transplanting, resulting in high post-planting survival rates. These practices ensured that the seedlings were robust and site-ready, significantly contributing to the successful establishment of the new plots.

- Species selection was informed by a careful analysis of environmental factors—a skill honed during the Endana training. At Kalpitiya, students assessed coastal salinity, tidal patterns, soil texture, and exposure to wind and sunlight to choose ecologically suitable species. This methodical, site-specific approach led to better plant establishment, minimized mortality, and promoted long-term sustainability.
- Perhaps the most impactful outcome of this field-based learning experience was the transformation of students from guided learners into confident leaders. In Endana, activities were supervised and instructional; in Kalpitiya, students led tasks such as establishing plots, collecting data, managing nurseries, and analyzing carbon sequestration. This empowerment underscores the value of experiential education in developing leadership, environmental responsibility, and scientific thinking among youth. The Kalpitiya project not only succeeded ecologically but also served as a model for student-led conservation, offering a replicable approach for other regions across Sri Lanka.
- A notable addition to the monitoring process was the assessment of tree condition within each plot. Using a standardized scoring method, the team calculated average condition values for both newly planted and existing mangrove vegetation. The average growth condition of newly planted mangroves was recorded at 0.885, while existing mangroves showed a slightly higher value of 0.894. This close margin indicates that the newly introduced vegetation is performing well and adapting effectively to the site. Although mature vegetation retains a marginally better overall condition, the strong performance of newly planted trees reflects effective species selection, site compatibility, and proper care. These findings are encouraging, suggesting that restoration efforts are narrowing the ecological gap between disturbed and mature zones, and reinforcing the project's long-term potential.
- In summary, the Kalpitiya Mangrove Restoration Project showcases the power of education, field training, and community engagement in driving meaningful ecological restoration. Through skillful application of methods learned in Endana, students demonstrated that with the right knowledge and tools, young people can lead scientifically sound and environmentally impactful projects. This experience not only restored a critical coastal ecosystem but also laid the groundwork for future student-led initiatives in biodiversity conservation across Sri Lanka.

5. Conclusions

The combined experience of the Endana and Kalpitiya projects illustrates how field-based ecological education can translate into impactful, student-led restoration. Skills gained in Endana were successfully applied in Kalpitiya, resulting in the establishment of monitored mangrove plots, creation of a digital inventory, and calculation of CO_2 sequestration (10.65 tCO_2/ha/yr). This initiative not only empowered students to take leadership roles but also demonstrated the potential of community-driven, scientifically grounded restoration to support national conservation goals and climate resilience. Looking ahead, the project serves as a scalable model for future ecological restoration efforts.

6. References

Komiyama, A., Ong, J. E., & Poungparn, S. (2005). Common allometric equations for estimating the tree weight of mangroves. *Journal of Tropical Ecology*, 21(4), 471–480. https://doi.org/10.1017/S0266467405002476

Defending mangroves and afforesting underwater. (2023, June 18). Dilmah Conservation. Retrieved April 18, 2025, from https://www.dilmahconservation.org/dc-news/defendingmangroves-and-afforesting-underwater--635n.html#

Mangrove ecosystems and livelihoods. (n.d.). Commonwealth. Retrieved April 19, 2025, from https://thecommonwealth.org/bluecharter/mangrove-restoration#:~:text=Mangroves%20provide%20a%20safe%20nursery,which%20flourish%20among%20their%20branches

Mangrove forests in worldwide decline. (2022, June 27). IUCN. Retrieved April 19, 2025, from https://iucn.org/content/mangrove-forests-worldwide-decline

Mangrove regeneration in Sri Lanka recognized as one of seven UN World Restoration flagships. (2024, February 16). United Nations Sri Lanka. Retrieved April 19, 2025, from https://srilanka.un.org/en/262045-mangrove-regeneration-sri-lanka-recognized-one-seven-unworld-restoration-flagships

United Nations Environment Programme. (2023, July 25). An inside look at the beauty and benefits of mangroves. UNEP. Retrieved April 19, 2025, from https://www.unep.org/news-andstories/story/inside-look-beauty-and-benefits-mangroves

7. Annexures

Kalpitiya Mangrove Inventory & Analysis Excel – GitHub Repository:
https://github.com/hasitha1021/Kalpitiya-mangrove-data-Analysis

Kalpitiya Mangrove Digital Inventory (not analysis)-GitHub Respository:
https://github.com/hasitha1021/Kalpitiya-MAngrove-Digital-Inventory

Endana Project Report –GitHub Repository:

https://github.com/hasitha1021/Dilmah_Endana-Project-Report